JourneyFileReader

+ parsePassengerType(String input): String
+ parseTimeBand(String input): String
+ readStudentsFromFile(String fileName):List<JourneyDetails>

JourneyFileWriter

+ writeStudentsToFile(String fileName, List<JourneyDetails> journeyList): void

1- Writes 1-- *

1- Reads 1-- *

JourneySavingsCalculator

- List <JourneyDetails> journeyDetails

+ setSavings(): void

1-- * Calculates 1-

JourneyDetails

- journeyID: int
- fromZone: int
- toZone: int
- timeBand: String
- passengerType: String
- journeyDate: LocalDate
- zonesCrossed: int
- basefare: BigDecimal
- discountedFare: BigDecimal
- chargedFare: BigDecimal

+ setDiscountedFare(BigDecimal discountedFare): void
+ setChargedFare(BigDecimal chargedFare): void
+ SetBaseFare(BigDecimal baseFare): void
+ getBaseFare(): BigDecimal
+ getDiscountedFare(): BigDecimal
+ getJourneyId(): int
+ getFromZone(): int
+ getToZone(): int
+ getTimeBand(): String
+ getPassengerType(): String
+ getChargedFare(): BigDecimal
+ getJourneyDate(): LocalDate
+ displayDetails(): void

JourneyManager

- List <JourneyDetails> journeyDetails
- totals : JourneyTotalByPassengerType
- lastRemovedJourney: JourneyDetails

+ addJourney (JourneyDetails journeyDetails, JourneyPricingRules journeyPricingRules): void
+ journeyIdExists (int id): boolean
+ calculateDisountForJourney(JourneyDetails journeyDetails): void
+ applyDiscount(JourneyDetails journeyDetails, BigDecimal baseFare): BigDecimal
+ displayJourneys(): void
+ filterByPassengerType(String type): void
+ filterByTimeBand(String timeBand): void
+ filterByStartingZone(int startingZone): void
+ filterByDestinationZone(int destinationZone): void
+ filterByJourneyDate(LocalDate date): void
+ removeJourneyById(int id, JourneyPricingRules journeyPricingRules): JourneyDetails
+ getLastRremovedJourney(): JourneyDetails
+ clearLastRemovedJourney(): void
+ undoLastRemoval(JourneyPricingRules journeyPricingRules): void
+ sortJourneysByFare(): void
+ displayDailySummary(): void
+ displayTimeBandCounts(): void
+ displayZoneCounts(): void
+ resetJourneys (boolean confirmed): void

1- Manages 1-- *

JourneyPricingRules

- List <JourneyDetails> journeyDetails
+ TimeBand: enum
+ PassengerType: enum

+ money(String amount): BigDecimal
+ key(int fromZone, int toZone, TimeBand timeBand): BigDecimal
+ getBaseFare(int fromZone, int toZone, TimeBand timeBand): void
+ buildBaseFare(): Map<String, BigDecimal>
+ put(Map<String, BigDecimal> m, int from, int to, TimeBand band, String amount): void
+ setBaseFareRules(): void
+ getDailyCap(String passengerType): BigDecimal
+ setDailyCap(String passengerType): BigDecimal
+ setAllJourneysDiscounts(): void

1-- * modifies 1-

Admin

- userName: String
- password: String
- baseFare: BaseFare

+ login(String inputUserName, String inputPassword): boolean
+ modifyBaseFareRules(): void
+ modifyAllJourneyDiscounts(): void
+ modifyAllJourneyCaps(JourneyTotalByPassengerType journeyTotalByPassengerType): void

1- might have 1-

CityRideLiteApp

journeyID: int
fromZone: int
toZone: int
passengerType: String
timeBand: String
journeyDate: LocalDate
baseFare: double

+ addJourney(journey, journeyPricingRules): void
+ journeyIdExists(journeyId): boolean
+ displayTotalsByPassengerType(): void
+ displayjourneys(): void
+ removeJourneyById(sc.nextInt(), journeyPricingRules): void
+ undoLastRemoval(removedJourney, journeyPricingRules): void
+ filterByPassengerType(sc.nextLine()): void
+ filterByTimeBand(sc.nextLine()):void
+ filterByStartingZone(fromZone): void
+ filterByDestinationZone(toZone): void
+ filterByJourneyDate(dateOfJourney): void
+ sortJourneysByFare(): void
+ displayDailySummary(): void
+ displayTimeBandCounts(): void
+ displayZoneCounts(): void
+ resetJourneys(sc.nextLine().equalsIgnoreCase("y")): void
+ recalculateTotals(journeyPricingRules): void
+ displayTotalsByPassengerType(): void
+ setTotals(journeyPricingRules): void
+ displayTotalsByPassngerType(): void
+ writeStudentsToFile("journeys.csv", journeyList): void
+ readStudentsFromFile("journeys.csv"): List<JourneyDetails>
+ calculateDisountForJourney(j): void
+ saveProfiles(): void
+ loadProfiles(): List <Profile>
+ login(inputUserName, inputPassword): void
+ setSavings(): void

1- adjusts 1-- *

JourneyTotalByPassengerType

- List <JourneyDetails> journeyDetails
- adultTotal: BigDecimal
- studentTotal: BigDecimal
- childTotal: BigDecimal
- seniorTotal: BigDecimal

+ recalculateTotals(GetCaps getCaps): void
+ setTotals(SetCaps setCaps): void
+ getRunningTotal(String passengerType): BigDecimal
+ addToRunningTotal(String passengerType, BigDecimal fare): void
+ displayTotalsByPassengerType(): void
+ resetTotals(): void

Rider

- List <Profile> profiles

+ addProfile(): void
+ displayProfiles(): void

1- might have 1-

1 has 1-- *

Profile

- name: String
- defaultPayment: String

+ showProfile(): void

1- needs 1-- *

ProfileFileReader

- gson: Gson
- fileName: String

+ loadProfiles(): List<Profile>

1- needs 1-- *

ProfileFileWriter

- List<Profile> profiles
- gson: Gson
- fileName: String

+ saveProfiles(): void